



ABBOTSLEIGH

Student Number: _____

Student Name : _____

Student Teacher: _____

AUGUST 2011

YEAR 12

ASSESSMENT 4

TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics

General Instructions

- Reading time – 5 minutes.
- Working time – 3 hours.
- Write using blue or black pen.
- Board-approved calculators may be used.
- A table of standard integrals is provided.
- All necessary working should be shown in every question.

Total marks (120)

- Attempt Questions 1-10.
- All questions are of equal value.

Question 1 (12 marks) Start a new booklet

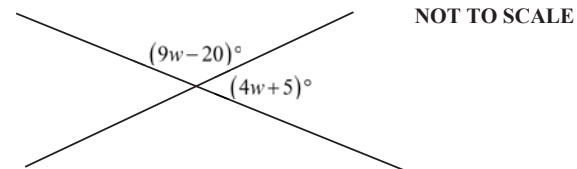
Marks

(a) Factorise $2h^2 + 11h + 15$

1

(b) Find the value of w in the diagram:

2



(c) Given $a = \frac{2}{7}$, $b = \frac{3}{5}$ and $c = 4\frac{1}{8}$, evaluate $\frac{b^2 - a}{2\sqrt{c}}$ in scientific notation to 3 significant figures.

2

(d) Express $\frac{\log_3 8}{\log_3 2}$ as an integer

1

(e) Determine the value of n to make the following expression equal to a single digit number:

1

$$5^2 \times 2^4 \times 10^{-n}$$

(f) Evaluate $\lim_{x \rightarrow 4} \frac{x^3 - 64}{x - 4}$

2

(g) Show that $\frac{21}{\sqrt{63}} - \frac{3}{\sqrt{7} + 2} = 2$

3

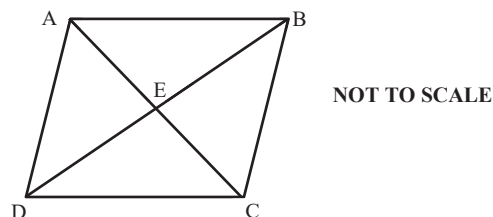
End of Question 1

Question 2 (12 marks) Start a new booklet

Marks

- (a) Find the area of the rhombus ABCD given $AB = 13$ cm and $EB = 12$ cm.

2



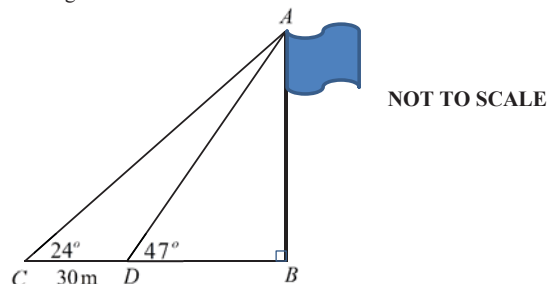
- (b) (i) Find the point A which is the y intercept of $4x - 3y - 12 = 0$

1

- (ii) Hence find the equation of the line passing through A, which is perpendicular to $4x - 3y - 12 = 0$

2

- (c) Harry finds that the angle of elevation of the top of a flagpole, A from C is 24° as shown in the diagram below. He walks 30 metres towards the flagpole and now finds that the angle of elevation is 47° .



- (i) Use the Sine rule to show that $AD = \frac{30 \sin 24^\circ}{\sin 23^\circ}$

2

- (ii) Hence show that $AB = \frac{30 \sin 24^\circ \sin 47^\circ}{\sin 23^\circ}$

2

- (iii) Calculate the length AB correct to 2 significant figures.

1

- (d) Solve for x : $(4x - 3)^2 = 25$

2

End of Question 2

Question 3 (12 marks) Start a new booklet

Marks

- (a) The points A and B have coordinates (2, 0) and (0, -2) respectively. Draw a diagram in your assessment booklet, clearly marking A and B.

- (i) Find the gradient AB.

1

- (ii) Show the equation of line l that passes through B and is perpendicular to AB is given by $x + y = -2$

2

- (iii) Show that C, the point of intersection of l and the x-axis has coordinates (-2, 0).

1

- (iv) If D is the point (0, 2), write down the equation of the circle passing through the points A, B, C and D.

1

- (v) Show the area between the circle ABCD and the quadrilateral ABCD is $4(\pi - 2)$ square units.

2

- (b) Calculate the perpendicular distance of the point (3, -1) from the line $4y = 3x + 2$.

2

- (c) Find the equation of the tangent to the curve $y = 5 \log_e x$ at $x = 1$.

3

End of Question 3

Question 4 (12 marks) Start a new booklet**Marks**(a) Differentiate with respect to x :

(i) $e^{\cos x}$

2

(ii) $\frac{2-x}{3x+4}$

2(b) Find the values of x for which the curve $y = x^3 - 6x^2 + 9x - 4$ is increasing.**2**

(c) The first three terms of an arithmetic progression are 51, 44 and 37.

(i) Write down the n th term for this sequence**2**

(ii) If the last term of the sequence is -47, how many terms are there in this series?

2

(iii) Find the sum of this series.

2**End of Question 4****Question 5 (12 marks) Start a new booklet****Marks**(a) If α and β are the roots of the equation $6x^2 - 2x + 1 = 0$ find:

(i) $\alpha + \beta$

1

(ii) $\alpha^2 + \beta^2$

2(b) Given the equation of the parabola $x^2 - 2x - 8y - 15 = 0$ (i) Show that the equation of the parabola can be expressed as:
 $(x-1)^2 = 8(y+2)$ **2**

(ii) Find the vertex.

1

(iii) Find the focus.

1

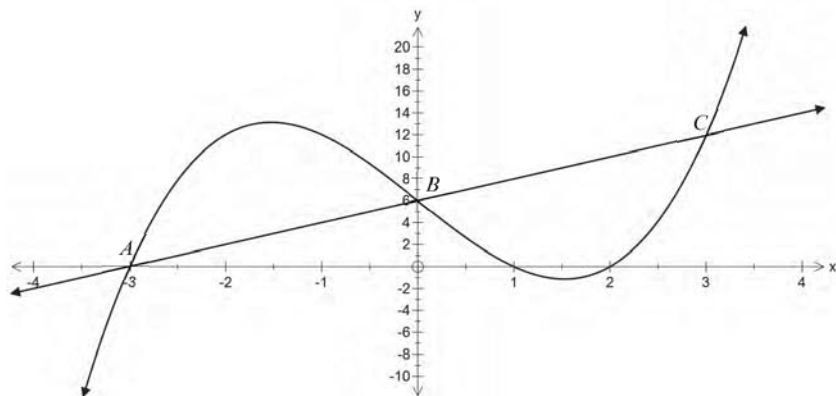
(iv) Find the equation of the directrix.

1(v) Sketch the parabola showing where it crosses the y axis, the focus and the directrix.**2**(c) The derivative of a function is given by $f'(x) = 15(5x-1)^2$.
If $f(0) = 10$, find the equation of $f(x)$.**2****End of Question 5**

Question 6 (12 marks) Start a new booklet

Marks

- (a) The following diagram shows the graphs of $y = x^3 - 7x + 6$ and $y = 2x + 6$



- (i) Find the coordinates of the intersection points A , B and C .

1

- (ii) Hence determine the domain of x such that:

2

$$x^3 - 7x + 6 > 2x + 6.$$

- (b) Find $\int \sqrt{5x-2} \, dx$

2

- (c) Evaluate the definite integral $\int_{\frac{\pi}{4}}^{\pi} 3 \sin 2x \, dx$

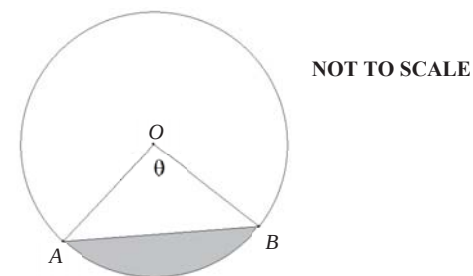
2

Question 6 continued on page 8

Question 6 continued

Marks

- (d) A circle has a radius of 20 cm and arc AB is 32 cm.



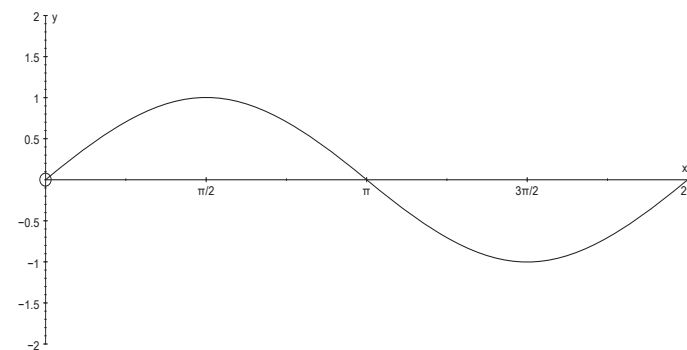
- (i) Find θ , the angle subtended at the centre by chord AB .
Give your answer in radians to 1 decimal place.

1

- (ii) Hence find the shaded area.

2

- (e) The diagram shows the graph of $y = \sin x$ in the domain $0 \leq x \leq 2\pi$.



- (i) Copy this graph into your assessment booklet **AND on the same set of axes** draw the graph of $y = \cos 2x$ in the domain $0 \leq x \leq 2\pi$.

1

- (ii) How many solutions are there to the equation $\sin x = \cos 2x$ in the domain $0 \leq x \leq 2\pi$?

1

End of Question 6

Question 7 (12 marks) Start a new booklet**Marks**

- (a) A geometric sequence has a second term 6 and the ratio of the sixth term to the fifth term is 3. Find the first term.
- (b) (i) Show that $\int_0^3 \frac{2}{x+1} dx = \log_e 16$
- (ii) Hence use the Trapezoidal rule with four function values to find an approximation of $\log_e 16$
- (c) Nancy's parents invest \$1200 each year in a superannuation fund for her. Compound interest is paid at 9% per annum on the investment. The first \$1200 is to be invested on Nancy's first birthday. The last is to be invested on her 21st birthday. To the nearest dollar:
- (i) How much is the first investment worth on Nancy's 22nd birthday?
- (ii) What is the total investment worth on Nancy's 22nd birthday?

2**2****3****2****3****End of Question 7****Question 8 (12 marks) Start a new booklet****Marks**

This question considers the function defined as $f(x) = e^{-\frac{x^2}{2}}$.

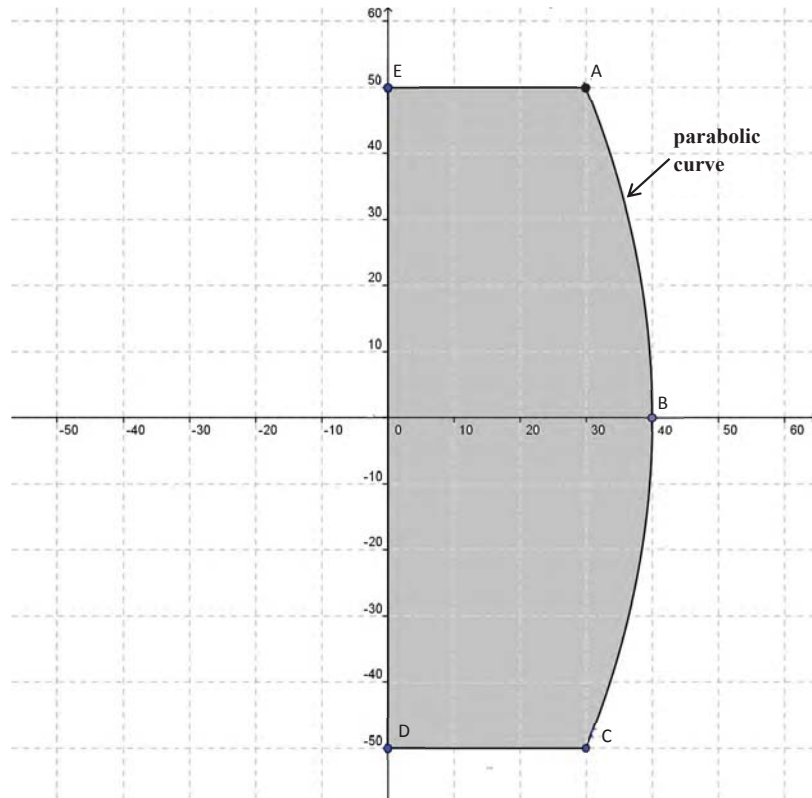
- (i) State the domain of $f(x)$.
- (ii) Show that $f(x)$ is an even function.
- (iii) Show that $f'(x) = -xe^{-\frac{x^2}{2}}$
- (iv) Find the stationary point of $y = f(x)$ and determine its nature.
- (v) Use the product rule to show that $f''(x) = (x^2 - 1)e^{-\frac{x^2}{2}}$
- (vi) Find the two points at which $f''(x) = 0$ and show they are points of inflexion.
- (vii) By considering the value that $f(x)$ approaches as x becomes large, state the range of $f(x)$.
- (viii) Sketch $y = f(x)$ showing the information found above.

1**1****1****2****2****2****1****2****End of Question 8**

Question 9 (12 marks) Start a new booklet

Marks

- (a) A wine barrel has been designed by rotating the shape ABCDE (shown in the following diagram) about the y axis. The curve ABC is parabolic. The point B is the vertex of this parabolic curve. All units on the graph are shown in cm.



- (i) Using the formulae $(y - k)^2 = -4a(x - h)$ show that the equation of the parabolic curve in the diagram is $y^2 = -250x + 10000$ **3**
- (ii) All units on the graph are shown in cm. By rotating the shaded area **around the y-axis**, find the volume of the barrel in Litres, where $1 \text{ cm}^3 = 1 \text{ mL}$. **3**

Question 9 continued on page 12

Question 9 continued

Marks

- (b) A particle is moving in a straight line. Its velocity, v as a function of time t ($t \geq 0$) is given by $v = \frac{4}{t+1} - 2t$.
- (i) Find when the particle changes direction. **2**
- (ii) Find the exact distance travelled in the first two seconds. **2**
- (iii) What is the acceleration of the particle as $t \rightarrow \infty$ **2**

End of Question 9

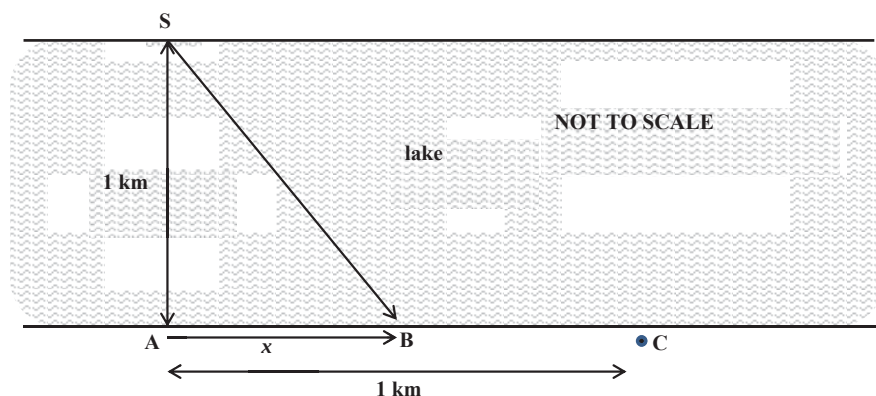
Question 10 (12 marks) Start a new booklet

Marks

(a) Show that $\frac{(1 + \tan^2 \theta) \cot \theta}{\operatorname{cosec}^2 \theta} = \tan \theta$ 3

(b) By using the identity $\cos \alpha = 2 \cos^2 \frac{\alpha}{2} - 1$, (do not prove this) 3
find the solutions to the equation $\cos \frac{\alpha}{2} = 1 + \cos \alpha$ for the domain $0 \leq \alpha \leq 2\pi$

- (c) Suzy wishes to return to camp. She is standing at S, on the edge of a lake, which is 1 km wide. The camp (at C) is one km from the direct opposite side (Point A) from where Suzy is currently standing, as shown in the diagram. She knows she walks at 3 km/h and swims at 2 km/h and wonders to herself at what distance, x , from the point opposite, should she swim to, in order to minimise the time to get to camp. Note that point B is a distance of x from point A.



(i) Using $\text{time} = \frac{\text{distance}}{\text{speed}}$, show that the total elapsed time, T in swimming to point B and walking from there to camp is given by $T = \frac{3\sqrt{1+x^2} + 2 - 2x}{6}$ 2

(ii) Knowing that Suzy wants to take the least amount of time getting back to camp, show that $x = \frac{2}{\sqrt{5}} \text{ km}$ AND determine her travel time in hours (correct to one decimal place). 4

End of assessment

Abbotsleigh Trial Mathematics Exam 2011

Question 1

(a) $(2h+5)(h+3)$

(b) $9w-20+4w+5=180$

$13w=195$

$w=15$

(c) $\frac{b^2-a}{2\sqrt{c}} = \frac{(\frac{3}{5})^2 - \frac{2}{7}}{2\sqrt{4\frac{1}{8}}}$

$\approx 0.01828787861...$

$\approx 1.83 \times 10^{-2}$

(d) $\log_3 8 = \frac{\log_3 2^3}{\log_3 2}$
 $= \frac{3\log_3 2}{\log_3 2}$
 $= 3$

(e) $5^2 \times 2^4 \times 10^{-n} = 400 \times 10^{-n}$
 $= 4 \times 10^2 \times 10^{-n}$

$\therefore 2-n=0$

$n=2$

(f) $\lim_{x \rightarrow 4} \frac{x^3-64}{x-4}$

$= \lim_{x \rightarrow 4} \frac{(x-4)(x^2+4x+16)}{x-4}$

$= 4^2 + 4 \times 4 + 16$

$= 48$

(1) (g) $\frac{21}{\sqrt{63}} - \frac{3}{\sqrt{7+2}} \times \frac{\sqrt{7-2}}{\sqrt{7-2}}$

$= \frac{21\sqrt{7}}{3\sqrt{7}\sqrt{7}} - \frac{3(\sqrt{7}-2)}{3}$

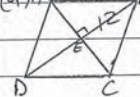
$= \frac{21\sqrt{7}}{21} - \sqrt{7} + 2$

$= \sqrt{7} - \sqrt{7} + 2$

$= 2$

Question 2

(a) A B AE=5 (by Pythagoras)



$A = \frac{1}{2} \times AC \times DB$

$= \frac{1}{2} \times (2 \times 5) \times (2 \times 12)$

$= 120 \text{ cm}^2$

(b) (i) A + A, $x=0$

$\therefore 0-3y-12=0$

$y=-4$

$A(0, -4)$

(ii) $3y = 4x - 12$

$y = \frac{4}{3}x - 4$

$\therefore m_1 = \frac{4}{3}, m_2 = -\frac{3}{4}$

eqn of line with m_2 & thru A

is $y = -\frac{3}{4}x - 4$

(c) (i) In $\triangle ACD$, $\angle C = 24^\circ$, $\angle D = 47^\circ$

$\frac{AD}{\sin 24^\circ} = \frac{30}{\sin \angle CAD = 23^\circ}$

$AD = \frac{30 \sin 24^\circ}{\sin 23^\circ}$

≈ 30.5

Q2 (c) (i)

In $\triangle ABD$, $\angle B = 90^\circ$

$\therefore \sin 47^\circ = \frac{AB}{AD}$

$AB = AD \sin 47^\circ$

$= \frac{30 \sin 24^\circ \sin 47^\circ}{\sin 23^\circ}$

(iii) $AB = 22.83936...$

$\approx 23 \text{ m to 2 sig figs}$

(d) $(4x-3)^2 = 25$

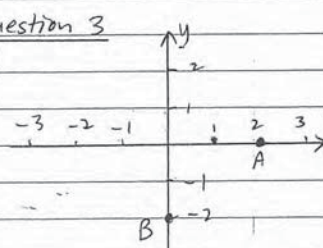
$4x-3 = \pm 5$

$4x = 5+3$ or $4x = -5+3$

$x = 2$ or $x = -\frac{1}{2}$

Question 3

(a)



(i) $m_{AB} = \frac{2}{2} = 1$

(ii) $m_L = -1$ $y_{\text{int}} = -2$

$\therefore y = -x - 2$

$x+y = -2$

(iii) L meets x axis when $y=0$

$\therefore C = (-2, 0)$ ($x+0=-2$)

(iv) circle has centre (0,0)

radius 2 $\therefore x^2 + y^2 = 4$

(v) $A = \pi r^2 - AB^2$

$= \pi \times 2^2 - (4+4)$

$= 4\pi - 8$

$= 4(\pi - 2) \text{ u}^2$

Q3(b) dist. of $(3, -1)$ from

$3x - 4y + 2 = 0$ is

$d = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$

$= \frac{|3 \times 3 - 4 \times (-1) + 2|}{\sqrt{9 + 16}}$

$= \frac{15}{5}$

$= 3 \text{ units}$

(c) $y = 5 \log_e x$

when $x=1$, $y = 5 \log_e 1$

$= 0$

$\frac{dy}{dx} = 5 \times \frac{1}{x}$

$A(1,0)$ $m=5$

$\therefore$ eqn of tangent is

$y - 0 = 5(x - 1)$

$y = 5x - 5$

Question 4

(a) (i) $y = e^{\cos x}$

$y' = -\sin x e^{\cos x}$

(ii) $y = \frac{2-x}{3x+4}$

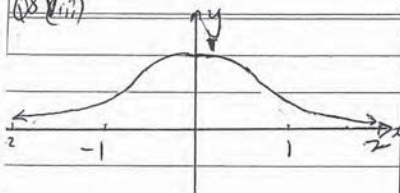
$y' = \frac{(3x+4)(-1) - (2-x)(3)}{(3x+4)^2}$

$= \frac{-3x-4-6+3x}{(3x+4)^2}$

$= \frac{-10}{(3x+4)^2}$

$= -\frac{10}{(3x+4)^2}$

Q8 (ii)



Question 9

(a) (i) vertex is $(40, 0) = B$ $\therefore h = 40 \quad k = 0$ So $y^2 = -4a(x - 40)$ At A, $x = 30, y = 50$ $\therefore 50^2 = -4a \times -10$ $a = \frac{2500}{40}$ $= \frac{125}{2}$ $\therefore y^2 = -250(x - 40)$ $y^2 = -250x + 10000$

(ii) $V = \pi \int_{-50}^{50} x^2 dy$

$= 2\pi \int_0^{50} \left(\frac{10000 - y^2}{250} \right)^2 dy$

$= 2\pi \int_0^{50} \frac{1600 - 8y^2 + y^4}{25 \times 62500} dy$

$= 2\pi \left[\frac{1600y}{25 \times 3} - \frac{8}{25 \times 3} \frac{y^3}{3} + \frac{y^5}{5 \times 62500} \right]$

$= 2\pi \left(\frac{1600 \times 50}{75} - \frac{8}{75} \times \frac{50^3}{3} + \frac{50^5}{312500} \right)$

$= \frac{406000\pi}{3}$

$\approx 425162.2058 \text{ cm}^3/\text{mL}$

$\approx 425.162 \text{ Litres}$

Question 8

(a) $f(x) = e^{\frac{x^2}{2}}$

(i) all real x

(ii) $f(-x) = e^{\frac{-x^2}{2}}$
 $= e^{-\frac{x^2}{2}}$
 $= f(x)$

 $\therefore$ even function

(iii) $f'(x) = \frac{-2x}{2} e^{-\frac{x^2}{2}}$
 $= -xe^{-\frac{x^2}{2}}$

(iv) stat. pt. when $f'(x) = 0$

$\therefore -xe^{-\frac{x^2}{2}} = 0$

only when $x = 0$ since $e^{-\frac{x^2}{2}} \neq 0$ At $(0, 1)$, $f'(x) = 0$ If $x = -0.5$, $f'(x) > 0$ If $x = 0.5$, $f'(x) < 0$ $\therefore$ max. t. pt at $(0, 1)$

(v) $f''(x) = e^{-\frac{x^2}{2}} \cdot -1 + -x \cdot -xe^{-\frac{x^2}{2}}$
 $= e^{-\frac{x^2}{2}} (x^2 - 1)$

(vi) $f''(x) = 0$ when $x^2 - 1 = 0$

$\therefore x = \pm 1$ ($e^{-\frac{x^2}{2}} > 0$ for all x)

When $x = -1.5$, $f''(-1.5) = 0.40$ When $x = -0.5$, $f''(-0.5) = -0.66$ $\therefore$ concavity changes $\therefore$ inflexion at $(-1, e^{-\frac{1}{2}})$ When $x = 0.5$, $f''(0.5) = -0.66$ When $x = 1.5$, $f''(1.5) = 0.40$ $\therefore$ concavity changes $\therefore$ inflexion at $(1, e^{-\frac{1}{2}})$ (vii) As $x \rightarrow \infty$, $f(x) \rightarrow 0$ $\therefore$ range is $0 < y \leq 1$

Q9 (b)

(iii) $\text{acc} = \frac{dv}{dt}$

$= -4(t+1)^2 - 2$

$= -4 - 2$

$\frac{(t+1)^2}{(t+1)^2}$

As $t \rightarrow \infty$, $\frac{-4}{(t+1)^2} \rightarrow 0$

$\therefore \text{acc} \rightarrow -2 \text{ units/s}^2$

Question 10

(a) LHS = $\frac{(1 + \tan^2 \theta) \cot \theta}{\csc^2 \theta}$

$= \sec^2 \theta \cdot \cot \theta \div \frac{1}{\sin^2 \theta}$

$= \frac{1}{\cos^2 \theta} \cdot \frac{\cos \theta}{\sin \theta} \times \frac{1}{\sin^2 \theta}$

$= \frac{\cos \theta}{\cos^2 \theta} \cdot \frac{\sin^2 \theta}{\sin \theta}$

$= \frac{\sin \theta}{\cos \theta}$

$= \tan \theta$

$= \text{RHS}$

(b) $\cos \frac{\alpha}{2} = 1 + \cos \alpha$

$= 1 + 2\cos^2 \frac{\alpha}{2} - 1$

$\therefore \cos \frac{\alpha}{2} = 2\cos^2 \frac{\alpha}{2}$

$2\cos^2 \frac{\alpha}{2} - \cos \frac{\alpha}{2} = 0$

$\cos \frac{\alpha}{2} (2\cos \frac{\alpha}{2} - 1) = 0$

$\therefore \cos \frac{\alpha}{2} = 0$ or $\cos \frac{\alpha}{2} = \frac{1}{2}$

$\frac{\alpha}{2} = \frac{\pi}{2}$ or $\frac{\alpha}{2} = \frac{\pi}{3}$ ($0 \leq \frac{\alpha}{2} \leq \pi$)

$\therefore \alpha = \pi$ or $\frac{2\pi}{3}$

Q9

(b) $v = \frac{4}{t+1} - 2t$

(i) changes direction when it stops $\therefore v = 0$

$\therefore 2t = \frac{4}{t+1}$

$2t^2 + 2t - 4 = 0$

$t^2 + t - 2 = 0$

$(t+2)(t-1) = 0$

$t = -2$ or $t = 1$

 $t > 0 \therefore$ only $t = 1$

So particle changes direction after 1 second.

(ii) $x = \int \frac{4}{t+1} - 2t dt$

$= 4 \ln(t+1) - t^2$

When $t = 0$, $x = 4 \ln 1 - 0 (= 0)$

When $t = 1$, $x = 4 \ln 2 - 1$

When $t = 2$, $x = 4 \ln 3 - 4$

Distance travelled

$= 4 \ln 2 - 1 + (4 \ln 2 - 1) - (4 \ln 3 - 4)$

$= 4 \ln 2 - 1 + 4 \ln 2 - 1 - 4 \ln 3 + 4$

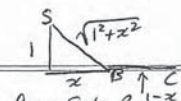
$= 8 \ln 2 - 4 \ln 3 + 2$

$= 2 + 4 \ln 4 - 4 \ln 3$

$= 2 + 4(\ln 4 - \ln 3)$

$= 2 + 4 \ln \frac{4}{3}$

10. (C) $T = D$



$\therefore$ time to swim from S to B

$$= \frac{\sqrt{1+x^2}}{2} \text{ hours}$$

$\therefore$ time to walk from B to C

$$= \frac{1-x}{3} \text{ hours}$$

(NB x is in km)

$$\therefore T = \frac{\sqrt{1+x^2}}{2} + \frac{1-x}{3}$$

$$= \frac{3\sqrt{1+x^2} + 2(1-x)}{6}$$

$$= \frac{3\sqrt{1+x^2} + 2 - 2x}{6} \text{ h}$$

(ii) T is minimum when $\frac{dT}{dx} = 0$

$$\frac{dT}{dx} = \frac{1}{6} \left(\frac{3}{2} (1+x^2)^{-\frac{1}{2}} \cdot 2x - 2 \right)$$

$$= \frac{x}{2\sqrt{1+x^2}} - \frac{1}{3}$$

$$= 0 \text{ when } \frac{1}{3} = \frac{x}{2\sqrt{1+x^2}}$$

$$\sqrt{1+x^2} = \frac{3x}{2}$$

$$1+x^2 = \frac{9x^2}{4}$$

$$4+4x^2 = 9x^2$$

$$x^2 = \frac{4}{5} \therefore x = \pm \frac{2}{\sqrt{5}}$$

but $x > 0 \therefore x = \frac{2}{\sqrt{5}} \text{ km}$

When $x = 0$, $\frac{dT}{dx} = -\frac{1}{3} < 0$

When $x = 1$, $\frac{dT}{dx} = 0 \dots > 0$

$\therefore$ min time when $x = \frac{2}{\sqrt{5}} \text{ km}$

$$\therefore T = \frac{3\sqrt{1+\frac{4}{5}} + 2 - \frac{4}{\sqrt{5}}}{6}$$

$$= 0.706 \dots \text{ h} \approx 0.7 \text{ h}$$